

AI-ML Avatars

What is Machine Learning?

What is Machine Learning?

A branch of Artificial Intelligence

- ▶ The design and development of algorithms
- ▶ Computers to capture and model behaviors
- ▶ Based on empirical data

What is Machine Learning?

Intelligence requires Knowledge

- ▶ It is necessary for the computers to acquire Knowledge
- ▶ Learn from external world; “teachers” etc. and solve problems
- ▶ Data provides knowledge in many cases

What is Machine Learning?

A very popular area now

- ▶ Lots of data
- ▶ Many recent success stories

What is Machine Learning?

Arthur Samuel, 1959

- ▶ Study that gives computers the ability to learn without being explicitly programmed

What is Machine Learning?

Arthur Samuel, 1959

- ▶ Study that gives computers the ability to learn without being explicitly programmed

Kevin Murphy, 2012

Algorithms that:-

- ▶ Automatically detect patterns in data
- ▶ Use the uncovered patterns to predict future data or other outcomes of interest

What is Machine Learning?

Tom Mitchell, 1997

Algorithms that:-

- ▶ Improve their performance (P)
- ▶ At some task (T)
- ▶ With experience (E)

What is Machine Learning?



Problem Space

Problem Space

- ▶ **Feature Extraction:** Find x corresponding to an entity/item (such as an image, web page, ECG etc.) Let us denote by I .

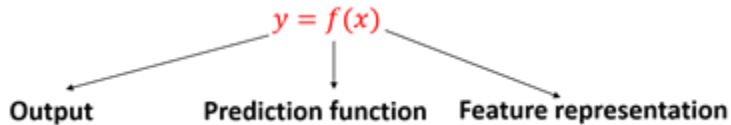
Problem Space

- ▶ **Feature Extraction:** Find x corresponding to an entity/item (such as an image, web page, ECG etc.) Let us denote by I .
- ▶ **Classification:** Find a parameterized function which can make the right predictions.
We denote the function by f .
We denote the predictions by y .

Problem Space

- ▶ **Feature Extraction:** Find x corresponding to an entity/item (such as an image, web page, ECG etc.) Let us denote by I .
- ▶ **Classification:** Find a parameterized function which can make the right predictions.
We denote the function by f .
We denote the predictions by y .
- ▶ **End to End:** Can we learn y directly from I ?

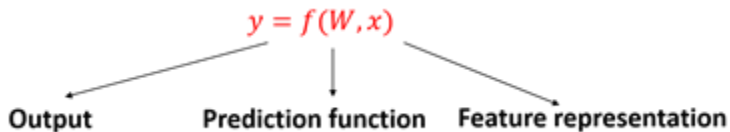
The Machine Learning Framework



Training: Given a training set, we estimate the prediction function, $f()$, by minimizing the prediction error.

Testing: Apply $f()$ to unknown test sample x and predicted value(output) is y .

The Machine Learning Framework



Training: Given a training set, we estimate the prediction function, $f(\cdot)$, by minimizing the prediction error.

Testing: Apply $f(W, \cdot)$ to unknown test sample x and predicted value(output) is y .

Parameter: Primary problem is to find the parameters, W

Spam Detection

Spam Detection



Spam Detection

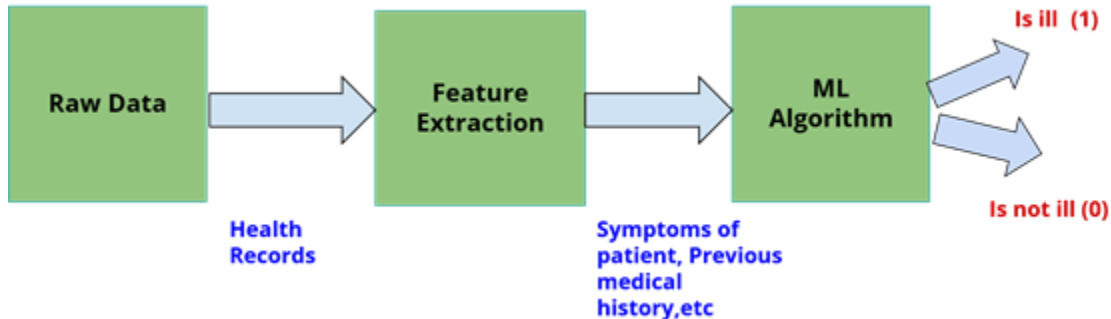


Medical Diagnosis

Medical Diagnosis

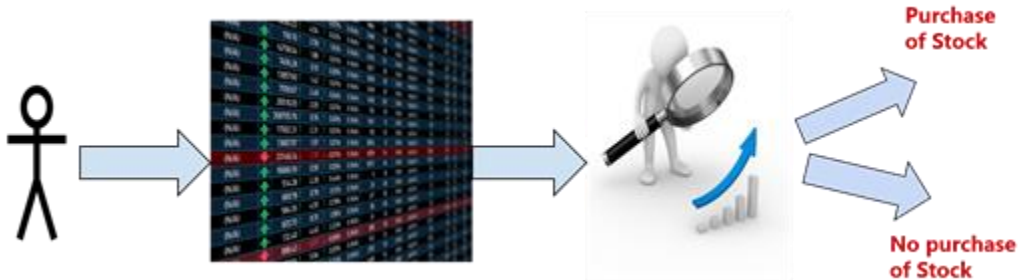


Medical Diagnosis

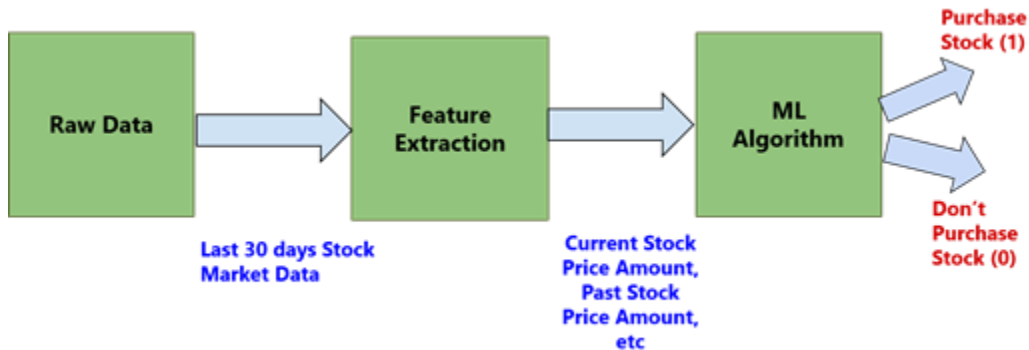


Stock Trading

Stock Trading

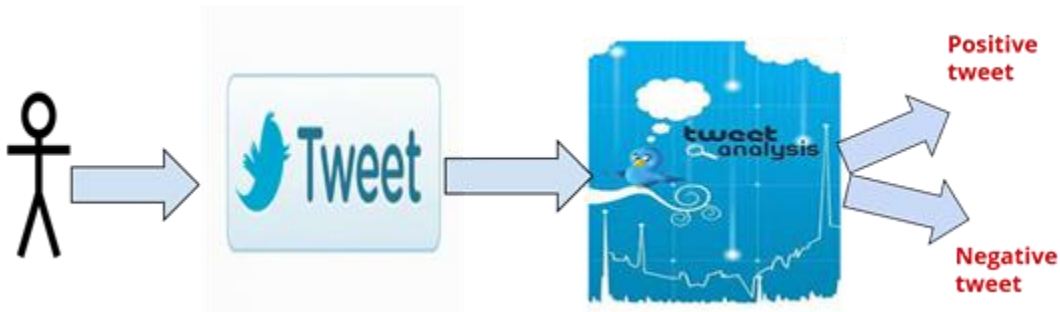


Stock Trading

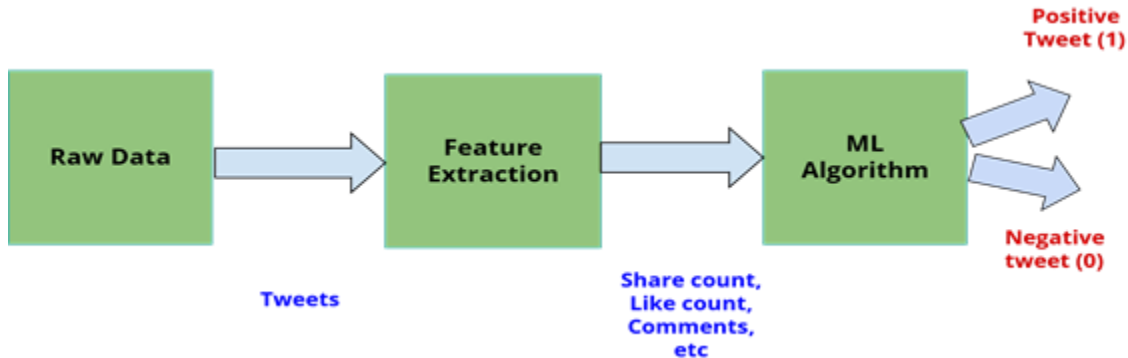


Sentiment Analysis

Sentiment Analysis

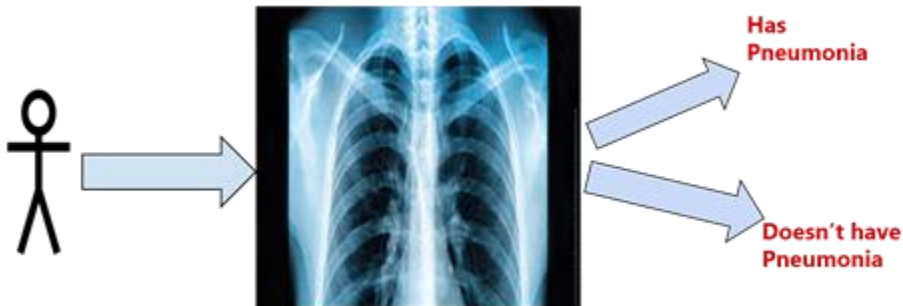


Sentiment Analysis

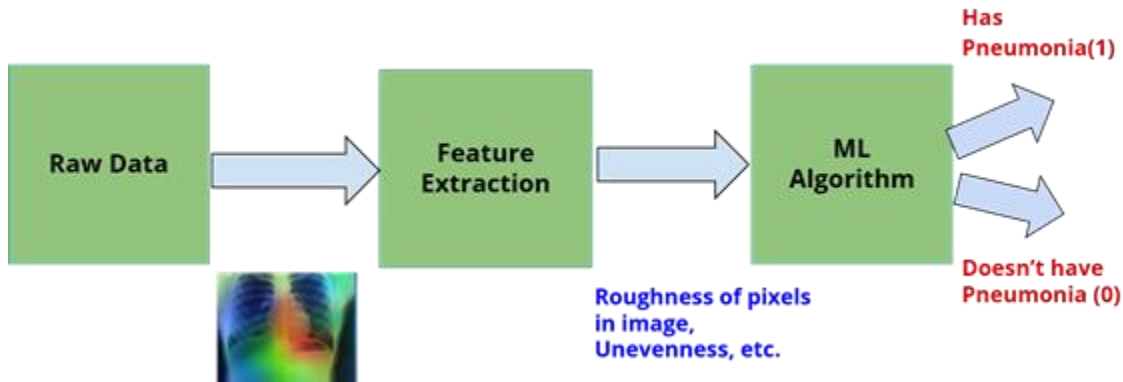


Disease Confirmation

Disease Confirmation



Disease Confirmation

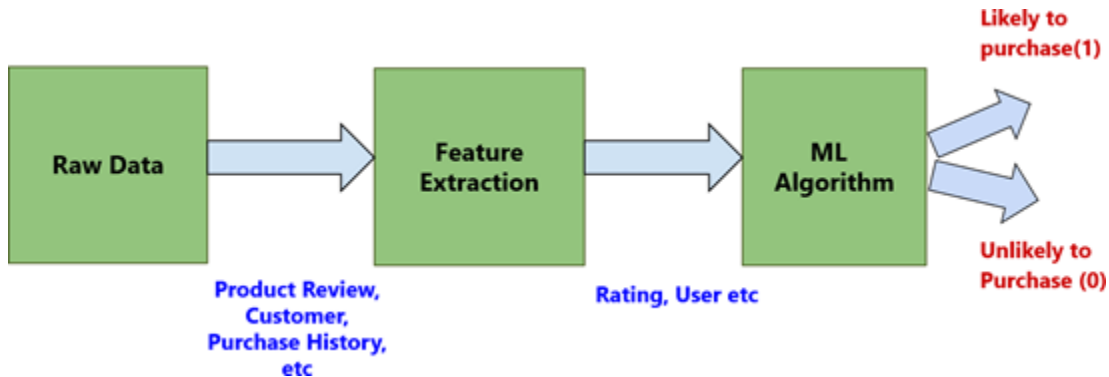


Product Recommendation

Product Recommendation



Product Recommendation

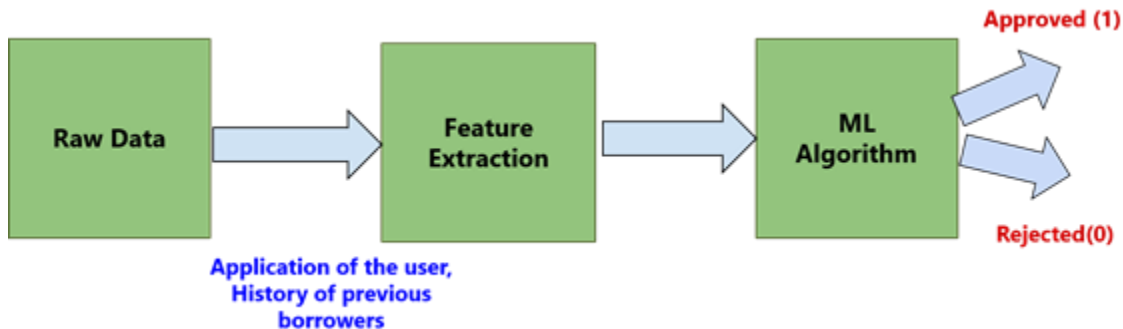


Loan Approval

Loan Approval



Loan Approval

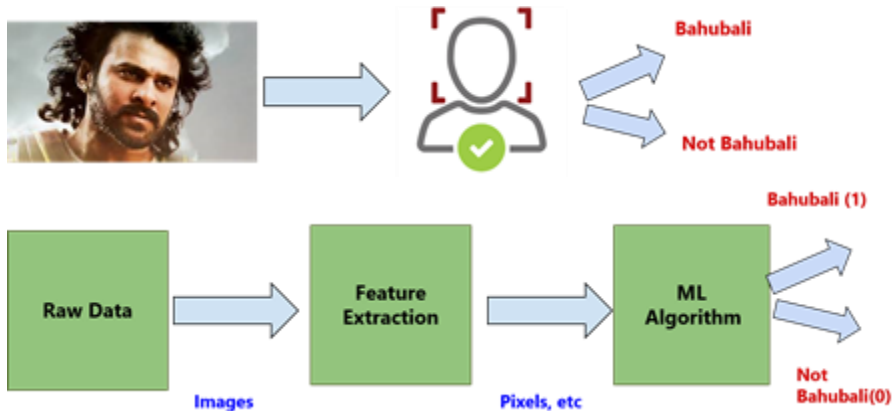


Face Recognition

Face Recognition

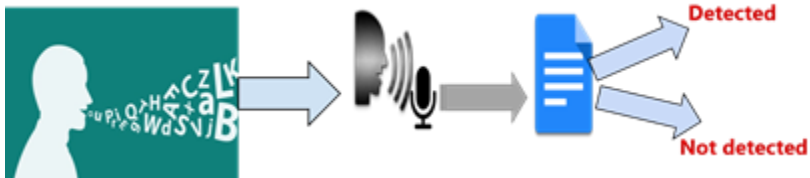


Face Recognition

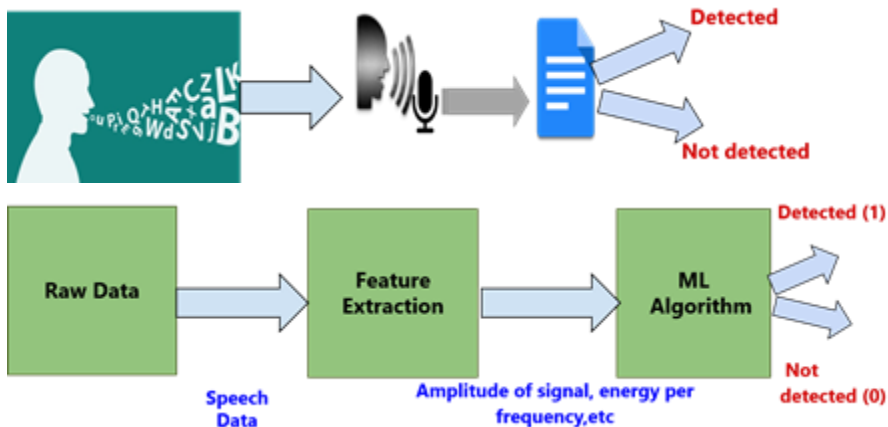


Voice Detection

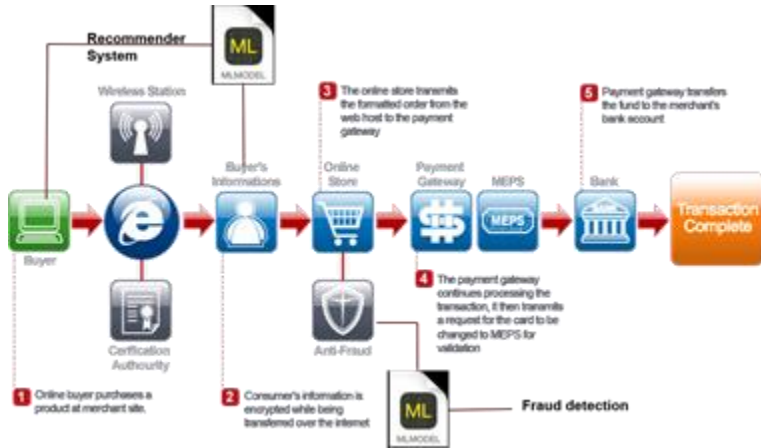
Voice Detection



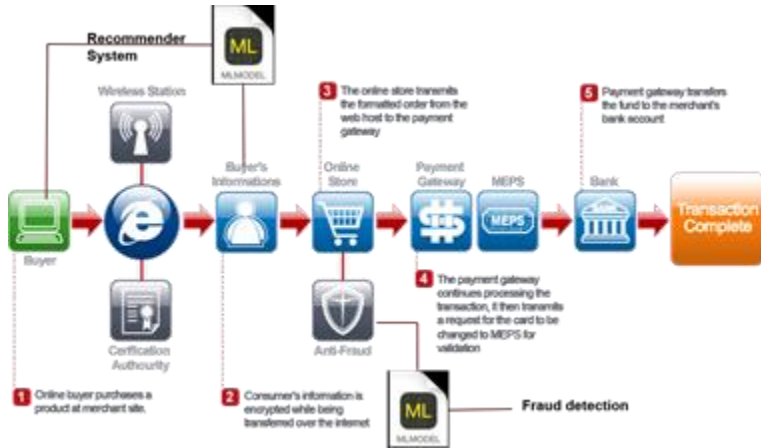
Voice Detection



ML's Role in a Large Software System



ML's Role in a Large Software System



ML can be considered as the **"intelligent" block** in a large software system

Thank you!
Questions?